CCEE Linux Exam Cheatsheet (Sessions 1–6)

System Architecture & Operation

- **Kernel:** Core of OS. Manages hardware (CPU, RAM).
- **Shell:** CLI that interprets user commands.
- **Run Levels:**
  - **0:** Halt/Shutdown
  - **1 (S):** Single-user (Root only, no network)
  - **3:** Multi-user, CLI only, Network enabled (Servers)
  - **5:** Multi-user, GUI enabled
  - **6:** Reboot

Essential Commands

Command	Function
`ls -l`	List files with details (permissions, owner, size).
`cp / mv / rm`	Copy / Move (Rename) / Remove.
`mkdir / rmdir`	Create dir / Remove **empty** dir.
`cat filename`	Display full file content.
`head / tail`	Show first/last 10 lines. (`tail -f` follows logs).
`grep "text" file`	Search for string inside a file.
`find / -name "x"`	Search for files in directory structure.
`tar -cvf`	Create Archive (No compression).
`tar -xvf`	Extract Archive.
`gzip / gunzip`	Compress / Decompress files.
`ps -ef`	Process Status (view running processes).
`kill <PID>`	Terminate a process by ID.

Permissions & Ownership

Permissions: Read (**r**=4), Write (**w**=2), Execute (**x**=1).

Numeric Mode (chmod)

Sum the values for **User**, **Group**, **Others**.

- `chmod 755 file`: User (4+2+1=7), Group (4+0+1=5), Others (5).
- `chmod 644 file`: User (rw-), Group (r–), Others (r–).

Symbolic Mode

u (user), g (group), o (others), a (all).
+ (add), - (remove). Example: `chmod u+x script.sh`

Ownership

- `chown user:group file`: Change owner and group.
- `chgrp group file`: Change group only.

Vi / Vim Editor

- **Command Mode:** Default on entry. Copy/Paste/Delete happen here.
- **Insert Mode:** Press `i` to type. Press `Esc` to exit.
- **Ex Mode:** Press `:` to save/quit.

Key	Action
`i`	Insert text before cursor.
`dd`	Delete (cut) current line.
`yy`	Copy (yank) current line.
`p`	Paste after cursor.
`:w`	Save file.
`:q!`	Quit **without** saving.
`:wq` / `:x`	Save and Quit.

Shell Scripting (Bash)

- **Shebang:** `#!/bin/bash` (First line).
- **Variables:** `count=10` (No spaces!). Access with `$count`.
- **Special Vars:**
  - `$0`: Script name.
  - `$1, $2...`: Command line arguments.
  - `$#`: Total number of arguments.
  - `$?`: Exit status of last command (0 = Success).

Redirection

- `>` : Overwrite output to file.
- `>>` : Append output to file.
- `|` (Pipe) : Output of Cmd1 becomes Input of Cmd2.

Logic

- `if [ $a -eq $b ]; then ... fi` (Spaces inside brackets vital).
- Comparison: `-eq, -ne, -gt, -lt` (numbers); `=, !=` (strings).